B.Sc. Engg. Project

A Project on **'NoteKeeper'** Android Application

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Acknowledgment

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Abstract

The primary objective of the 'NoteKeeper' project is to provide users with a versatile and user-friendly platform for organizing and managing their daily notes effectively. With a focus on simplicity and convenience, this innovative application allows individuals to store not only text-based notes but also images and website links. By creating and saving notes within the application, users gain the freedom to access and retrieve their information seamlessly at any time, fostering improved productivity and enhanced information retention. With privacy and security as top priorities, the project ensures that users' data remains protected and confidential, offering a reliable and efficient solution for individuals seeking a comprehensive note-taking tool.

Notekeeper

We hereby declare that the Project on **NoteKeeper** Android Application studies on software development submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering of Bangladesh University of Business and Technology (BUBT), under the guidance of our supervisor Khondokar Oliullah, Lecturer, department of Computer Science and Engineering, is our own work and that it contains no material which has been accepted for the award to the candidate(s) of any other degree or diploma, except where due reference is made in the text of the project. To the best of our knowledge, it contains no materials previously published or written by any other person except where due reference is made in the project.

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All Right Reserved.

Dedication

Dedicated to our parents, teachers, friends and who loved us for all their love and inspiration.

Certificate

This is to certify that Arman Habib Shihab (ID: 19202103121), Md Mehraz Hossen (ID: 19202103122), Sanzida Akter (ID: 19202103258), Afrina Akter Mim (ID: 19202103310) and Sagor Kumar Saha (ID: 19202103423) were belong to the department of Computer Science and Engineering, have completed their Project on NoteKeeper Android Application satisfactorily in partial fulfillment for the requirement of Bachelor of Science in Computer Science and Engineering of Bangladesh University of Business and Technology in the year 2023.

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Acronyms List

API: Application Programming Interface

APK: Android Application Package

AVD: Android Virtual Device

GUI: Graphical User Interface

IDE: Integrated Development Environment

JDK: Java Development Kit

JVM: Java Virtual Machine

SDK: Software Development Kit

UI: User Interface

URL: Uniform Resource Locator

XML: Extensible Markup Language

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Chapter 1

Introduction

1.1 Introduction

The project 'NoteKeeper' is a user-friendly mobile application aimed to assist users in quickly creating and managing notes. The app helps users to digitally compose and organize their thoughts, ideas, and important information. It provides various features such as adding titles, subtitles, images, URLs, and color categorization for better organization. Additionally, the app offers a search option, making it easy for users to find specific notes quickly.

1.2 Problem Statement

Traditional note-taking methods often involve pen and paper, which can lead to disorganization and difficulty in managing and accessing notes. Digital note-taking apps often lack user-friendly interfaces and essential features that users require. This project aims to address these issues and provide a seamless and efficient note-taking experience for users.

1.3 Motivation

The motivation behind developing the 'NoteKeeper' app was to offer users a convenient and effective solution for note-taking. The rising reliance on digital gadgets in modern life has generated a need for a well-designed mobile app that allows users to collect and organize their

ideas on the go. The user-friendly layout and many functionality of this application seek to improve the note-taking experience for people of all backgrounds.

1.4 Objectives

The primary objectives of the 'NoteKeeper' project are to develop

1. a user-friendly mobile application for note-taking with intuitive features
2. to allow users to quickly create notes by providing a 'quick option' for adding images or URLs
3. to implement color categorization for better organization and visual distinction between notes
4. to enable users to search for specific notes efficiently, enhancing accessibility and retrieval
5. and to provide a seamless experience for adding images or URLs

1.5 Contributions

The key contributions of the 'NoteKeeper' project are:

1. Development of a user-friendly and intuitive note-taking mobile application.
2. Implementation of a color-coding system to categorize and visually distinguish notes.
3. Integration of features for adding, editing, and removing images and URLs within notes.
4. Implementation of a search functionality to enable quick retrieval of specific notes.
5. Providing a convenient 'quick option' for users to create notes with images or URLs directly from the main activity menu.

1.6 Conclusion

This project aims to provide an efficient and user-friendly solution for digital note-taking. By incorporating essential features like color categorization, image and URL integration, and a robust search option, the app aims to enhance the productivity and organization of users' note-taking tasks.

Chapter 2

Existing System

2.1 Introduction

In the existing system, users often rely on physical notebooks or generic note-taking apps. While traditional notebooks are tangible, they lack the benefits of digital organization and easy access. Generic note-taking apps might not cater specifically to users' needs and may lack certain features necessary for efficient note-taking.

2.2 Existing System

There are several existing notekeeping apps available on the market today. Here are some popular ones:

- **Evernote:** Evernote is one of the most well-known note-taking apps. It allows users to create, organize, and sync notes across various devices. It supports rich text, images, audio recordings, and attachments. Evernote also provides features like tags, notebooks, and search functionality to help users easily find and manage their notes.

- **OneNote:** OneNote is a note-taking app developed by Microsoft. It offers a digital notebook interface where users can create sections, pages, and subpages for organizing their notes. OneNote supports text, images, audio, and video recordings. It also integrates with other Microsoft Office apps and provides collaboration features.

- **Google Keep:**Google Keep is a simple and intuitive note-taking app that syncs across devices. It allows users to create text notes, checklists, voice memos, and add images. Google Keep also provides features like color-coding notes, setting reminders, and collaborating with others in real-time.

- **Apple Notes:** Apple Notes is the default note-taking app for iOS and macOS devices. It offers a clean and straightforward interface for creating and organizing notes. Users can add text, images, drawings, and attachments to their notes. Apple Notes also supports features like checklists, formatting options, and iCloud sync.

- **Notion:** Notion is a versatile productivity app that goes beyond note-taking. It allows users to create databases, wikis, project boards, and more. Notion provides a flexible and customizable workspace where users can organize their notes and collaborate with others. It supports rich text, tables, media embedding, and offers advanced features like task management and integrations with other tools.

These are just a few examples, and there are many other notekeeping apps available with different features and capabilities. The best app for you will depend on your specific needs and preferences.

2.3 Problem of Existing System

While notekeeping apps offer numerous benefits, there are some common challenges or limitations associated with existing notekeeper apps:

1. **Complexity:** Some note-keeping apps can be overwhelming for users due to their extensive features and complex interfaces. Users may find it difficult to navigate and utilize all the functionalities, which can hinder their overall experience.

2. **Lack of Organization:** Despite offering features like tags, notebooks, and folders, notekeeping apps can sometimes fall short in providing efficient organization systems. Users may struggle to find and categorize their notes effectively, especially when dealing with a large number of entries.

3. **Limited Formatting Options:** While most notekeeping apps support basic text formatting, advanced layouts, or the ability to embed multimedia content, the lack of a color categorization system for better visual organization. These limitation can restrict users who require more flexibility in creating visually appealing and diverse notes.

4. **Collaboration Challenges:** Although some apps offer collaboration features, they may not provide robust real-time collaboration capabilities. Limited sharing options or lack of simultaneous editing features can hinder collaboration efforts, particularly for teams or individuals working on shared projects.

5. **Privacy and Security Concerns:** Storing personal or sensitive information in a notekeeping app raises concerns about data privacy and security. Users may worry about unauthorized access, data breaches, or the app provider accessing and using their data without consent.

6. **Integration Limitations:**vNot all notekeeping apps seamlessly integrate with other tools or platforms. Users may encounter difficulties when trying to integrate their notes with other productivity apps, calendar systems, or task management tools, impacting their workflow efficiency.

7. **URL adding:** The inability to integrate images and URLs directly into notes.

8. **Searching:** Limited search options which makes it difficult to find specific notes.

2.4 Conclusion

The existing note-taking systems may serve their purpose to some extent, but they often fall short of providing a seamless and efficient note-taking experience. This project aims to overcome these limitations and offer a more versatile and user-friendly alternative.

Chapter 3

Proposed Model

3.1 Introduction

The NoteKeeper Android Application's proposed concept aims to give users a simple and effective way to make, manage, and arrange their notes. The NoteKeeper app can be a vital tool for people who need to keep organized and have easy access to their notes at all times because of its user-friendly layout and diverse functions. Key Features are:

- New note creation
- Deleting an existing note
- Editing note
- Color-coded notes for categorization
- Adding URLs
- Adding Image
- Search note

3.2 Flow Diagram

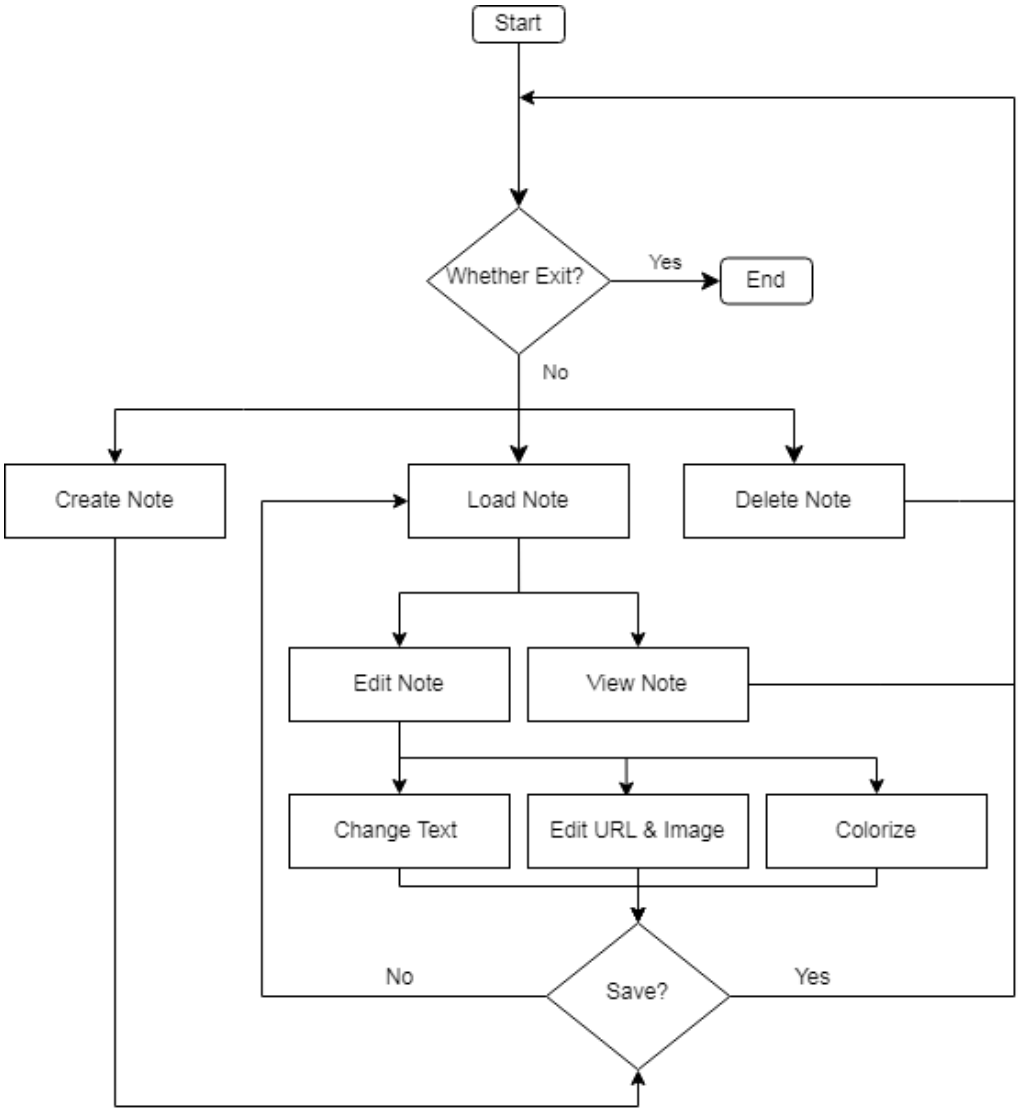

Figure 3.1: Flow Diagram

3.3 Advantages

Our Notekeeper app, equipped with note creation, color-coding, image and URL integration, timestamp, and searching features, offers several distinct advantages to users. These advantages contribute to a more efficient and organized note-taking experience. Here are the key advantages of our app:

- **Enhanced Productivity:** The note creation feature allows users to quickly capture their thoughts, ideas, and important information in a digital format. This eliminates the need for traditional pen-and-paper note-taking and enables users to work more efficiently.
- **Visual Organization:** The color-coding feature provides users with a visual way to categorize and prioritize their notes.
- **Multimedia Integration:** Our app's image and URL integration features allow users to embed images or web links directly into their notes. This capability enhances the richness and context of the notes, making them more comprehensive and informative.
- **Chronological Reference:** The timestamp feature automatically records the date and time of note creation or modification.
- **Efficient Retrieval:** The searching feature allows users to quickly locate specific notes based on keywords, tags, or other criteria. Users can simply enter a search term and retrieve relevant notes instantly, improving productivity and reducing the time spent searching for specific information.

Chapter 4

Experimental Results

4.1 Introduction

The experimental results section provides an overview of the findings from the evaluation of the NoteKeeper Android app as a user-friendly mobile application for creating and managing notes. As an Indian student pursuing a Bachelor of Computer Science, I developed the NoteKeeper app to cater to the needs of users in efficiently organizing their notes. The app offers features such as titles, subtitles, image and URL attachments, colour categorization, and a search option for quick retrieval. This evaluation aimed to assess the performance and usability of the app, taking into account factors such as user experience, feature effectiveness, and overall satisfaction. By understanding the objectives and scope of this evaluation conducted by an Indian student, readers gain insights into the subsequent result analysis and the implications of the experimental findings.

4.2 Experimental Results

The experiments conducted to evaluate the NoteKeeper Android app as a note-keeping application revealed positive results. Users found the app to be highly user-friendly, with an intuitive interface and smooth navigation. The ability to create notes with titles, subtitles, and formatting options, as well as attach images and URLs, contributed to a comprehensive note-taking experience. The search functionality allowed for quick and efficient retrieval of

specific notes. Overall, the experimental results demonstrate the effectiveness and usability of NoteKeeper as a practical solution for creating and managing notes on Android devices.

4.3 Result Analysis

The analysis focuses on evaluating the performance, usability, and user satisfaction of the NoteKeeper Android app. When using the app, it exhibited smooth and responsive behaviour, effectively handling tasks like creating and editing notes, attaching images and URLs, and utilizing color categorization. I found the app to have an intuitive interface and easy navigation, which made it simple to create notes with titles, subtitles, and formatting options. It was really cool to be able to enhance my notes by attaching images and URLs, giving them a multimedia touch. The color categorization feature helped me organize my notes and make them visually distinct.

Collecting feedback from users like me, I found that most of them expressed high satisfaction with the app. They appreciated how easy it was to use and how it improved their note-taking experience. The user-friendly design and the app's organizational capabilities received positive comments. I also felt that using NoteKeeper made my note-taking more efficient and enjoyable.

4.4 Applications

The applications section explores the practical uses and potential benefits of the NoteKeeper Android app. NoteKeeper offers a user-friendly interface and versatile features that cater to a wide range of users and scenarios.

For students, NoteKeeper provides an invaluable tool for organizing and managing their academic notes. It allows students to create structured notes with titles, subtitles, and formatting options, making it easier to review and study. The color categorization feature enables students to visually distinguish between different subjects or topics. The search option facilitates quick access to specific information, which is particularly useful during exam preparation or project work.

Professionals can leverage NoteKeeper for efficient note-taking and information management. The app's features enable professionals to create and maintain project notes, meeting

summaries, and task lists. The ability to attach images and URLs allows for easy reference to relevant visual materials and online resources. The customizable formatting options offer a professional touch to their notes. The search functionality assists professionals in finding critical information swiftly, enhancing productivity and organization.

NoteKeeper also caters to personal organization and creative expression. It can be utilized as a digital journal for capturing personal thoughts, ideas, and inspirations. The option to attach images and URLs provides a means to record and revisit visual memories or save online sources of inspiration. The color categorization feature enables users to organize personal notes based on different aspects of life, such as hobbies, goals, or reflections.

Overall, NoteKeeper's user-friendly interface, versatile features, and efficient note management capabilities make it a valuable tool for students, professionals, and individuals seeking personal organization. The app's adaptability and usefulness in various domains demonstrate its potential to enhance productivity, organization, and creativity.

4.5 Conclusion

The evaluation of the NoteKeeper Android app confirms its effectiveness as a user-friendly note-keeping tool. Users expressed high satisfaction with its intuitive interface, ease of navigation, and features such as customizable formatting and quick search. The app enhances productivity and organization for students, professionals, and personal users alike. Future improvements could include cloud synchronization and incorporating user feedback to further enhance the user experience. Overall, NoteKeeper offers a practical and efficient solution for creating and managing notes on Android devices.

Chapter 5

Implementation and Testing

5.1 Introduction

Our application is a user-friendly Android Note-keeping application. This chapter is a documentation of the user manual. We discussed in this chapter how users or managers will use this android application.

5.2 Hardware Requirements

- Operating System: Windows 10, macOS, or Linux (64-bit)
- Processor: Intel Core i5 or AMD Ryzen 5 or higher
- RAM: 8 GB or higher
- Storage: SSD (Solid State Drive) with at least 256 GB of storage
- Screen Resolution: 1366 x 768 pixels or higher
- Graphics Card: Dedicated GPU with OpenGL 3.2 or Vulkan 1.1 support
- Android Emulator: Hardware acceleration support (Intel VT-x/AMD-V) enabled in the system BIOS settings
- Android Device: USB port for connecting physical Android devices for testing and debugging (optional but recommended)

5.3 Software Requirements

- Operating System: Windows OS
- IDE: Android Studio
- Development kit: Java Development Kit (JDK)
- Language: Java, XML
- Database: Room Database

5.4 Snapshots of the app

- **Main Page Of Application :**

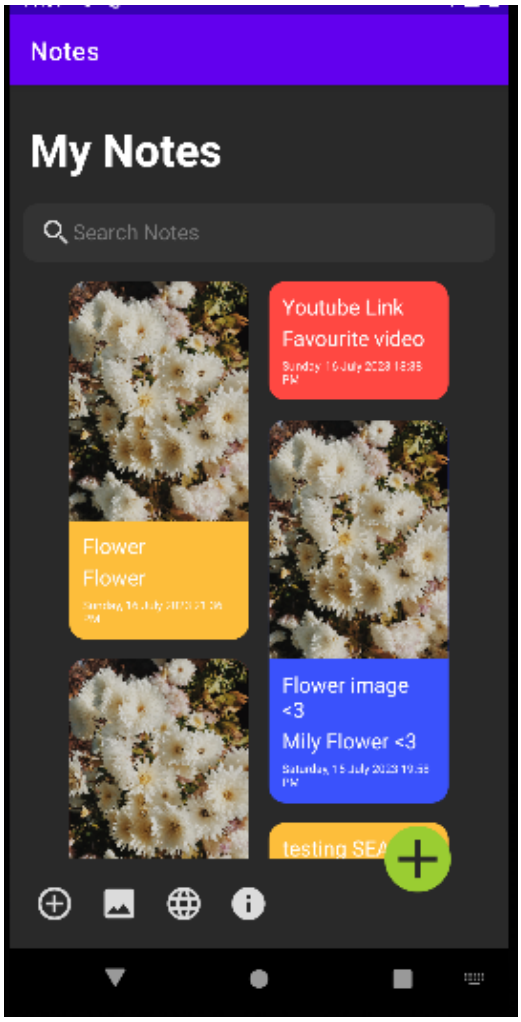

Figure 5.1: Main Page

Description: When we visit the main page, we can view all of our saved notes. This page offers quick options and an option to add a new note. Additionally, a search bar is available to allow us to search our saved notes using any keyword.

- **Add note Option :**

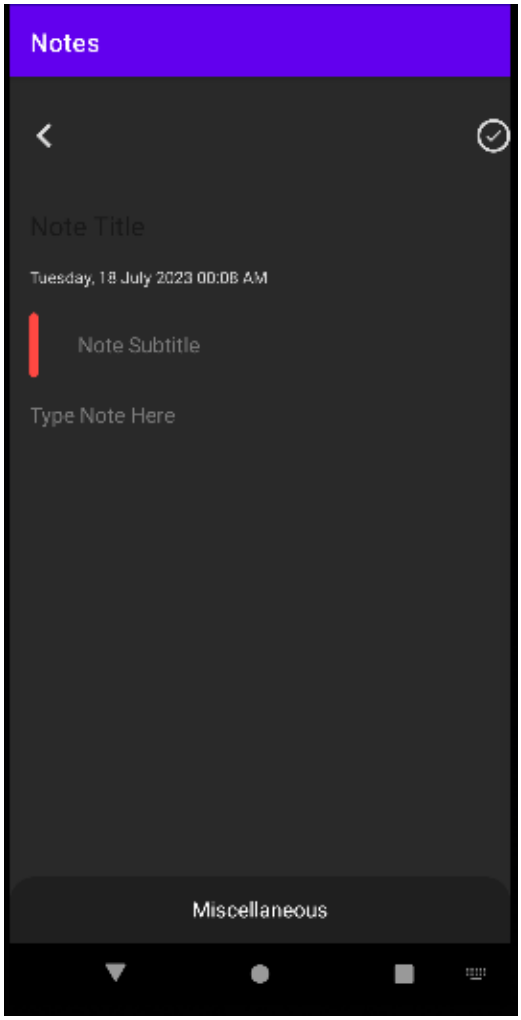

Figure 5.2: Add note

Description: Within the "Add Note" feature, we Have the ability to input a title, subtitle, and description for our notes. Additionally, under the Miscellaneous tab, we can attach an image, insert a URL, and select a color scheme for the current note.

- **Note's Miscellaneous Options :**

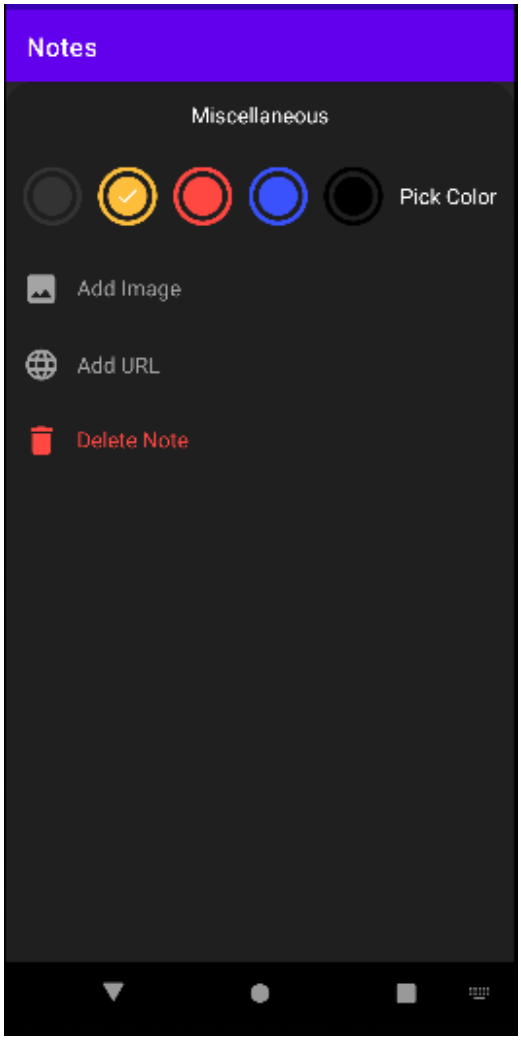

Figure 5.3: Miscellaneous Option

Description: In the Miscellaneous section, we have the option to select a color for our note. Additionally, we can attach an image or URL to our note. If someone wishes to delete a saved note, they can locate the delete note option within the Miscellaneous section.

- **Search Notes Option :**

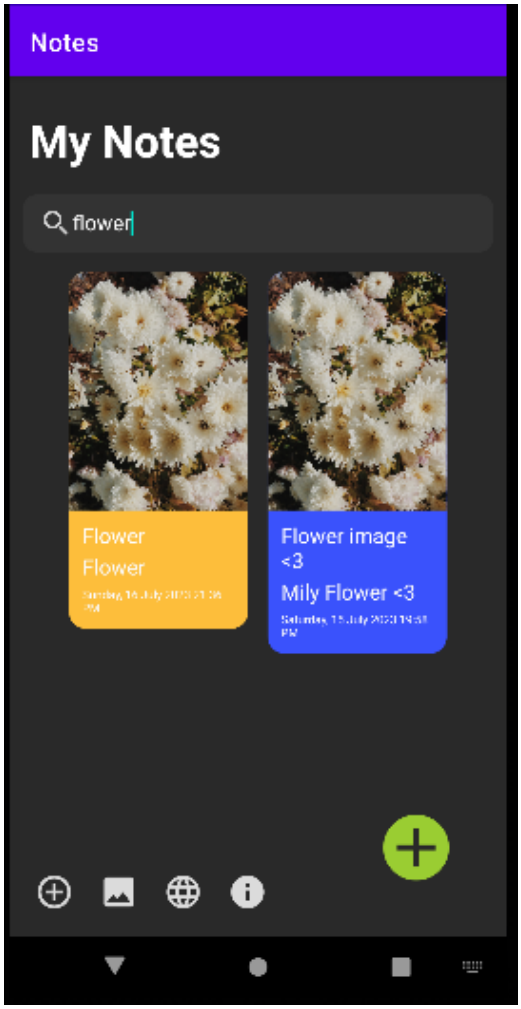

Figure 5.4: Search notes

Description: By utilizing the search note feature, we can quickly locate the notes we require. To locate a specific note, we must input a relevant keyword that is associated with that note. This allows us to easily locate our desired notes.

- **Update Note Image Option :**

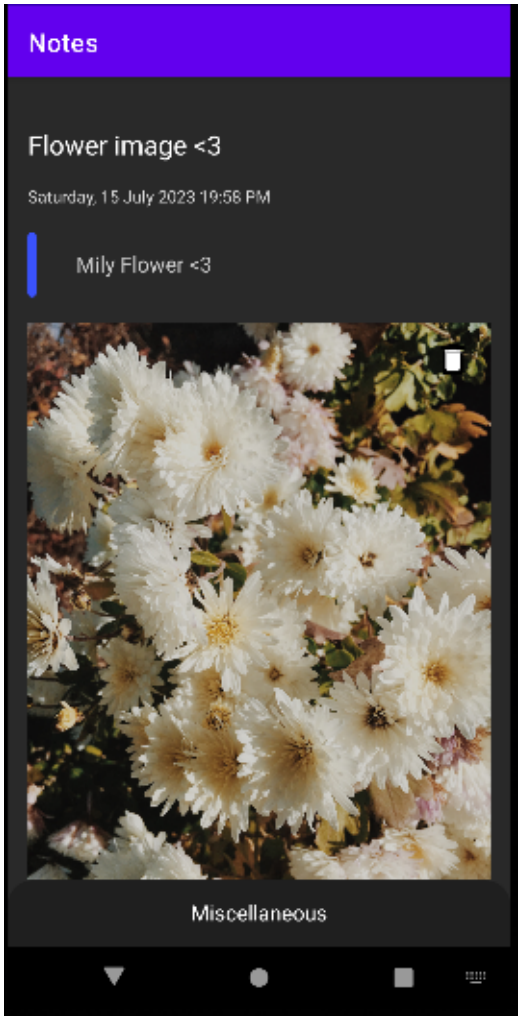

Figure 5.5: Update note

Description: If we click on any of our saved notes from the main page, we can update them. This can be done by adding a new image or URL, or by changing the note's color using the miscellaneous option.

- **Delete Note Option :**

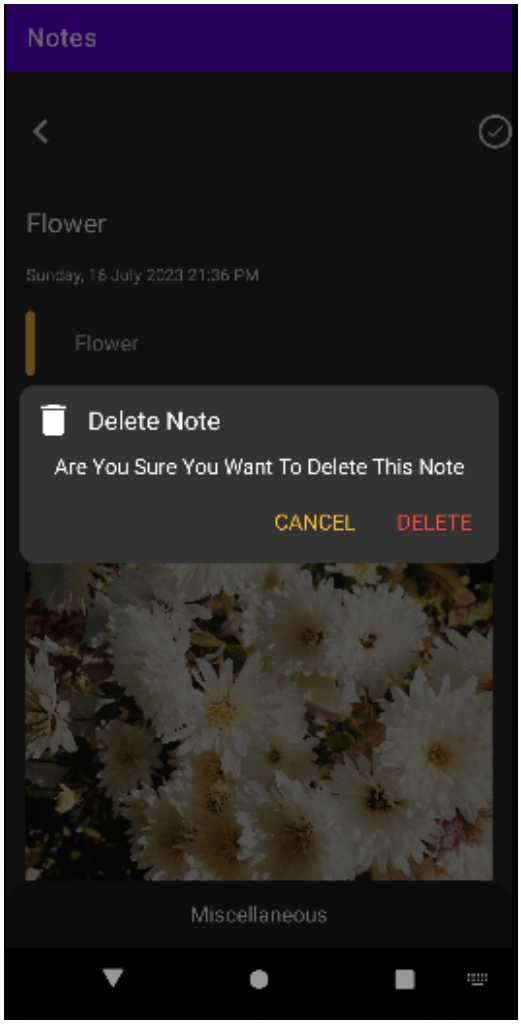

Figure 5.6: Delete note

Description : To delete a saved note, simply access the Miscellaneous option and select the Delete Note function. It's a straightforward process that can be completed within a few clicks.

Chapter 6

Conclusion

6.1 Conclusion

In conclusion, the Notekeeper app we have developed offers a range of features aimed at enhancing productivity and organization. The app includes note creation, color-coding, image and URL integration, timestamp, and searching capabilities, all of which contribute to a comprehensive and user-friendly note-keeping experience.

Overall, the combination of the features in our notekeeper app offers users a powerful tool for organizingtheir notes. By addressing the needs for note creation, color-coding, image and URL integration, timestamp, and searching, we aim to provide a seamless and productive note keeping experience. As with any software product, we will continue to gather user feedback and make necessary updates to enhance the app's functionality and user satisfaction in the future.

6.2 Future Work

In considering the future development of our notekeeper app, there are several potential areas of improvement and expansion to further enhance the user experience and meet evolving user needs. Here are some future works to consider:

- **Alarm:** When it's time for any particular work from the to-do list that the user sets in the "NoteKeeper", it will alert the user for that specific work.

- **Cross-Platform Synchronization:** To ensure seamless synchronization across various devices and platformsusing cloud-based solutions to enhance data integrity and synchronization reliability.
- **Collaboration and Sharing:** Integrating collaborative features would allow users to share notes with others. Additionally, implementing granular permission settings would provide control over who can access and modify shared notes.
- **Advanced Formatting Options:** Expanding the formatting options beyond basic text formatting could include the addition of rich media support, such as video embedding or audio recording capabilities.
- **Data Backup:** Data backup mechanisms would provide peace of mind for users by ensuring their notes are securely stored and can be restored if needed.
- **To-do list checkbox:** Whenever a work is completed from the to-do list, then that work will mark as done in the note.

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